10)

b) Aim:

To Implement classless IP subnetting cisco packet tracer and establish connection blw different subnetworks using appropriate IP addresses and subnet Masks.

Procedure:

Step1: ① Open cisco packet tracer

② Add the required devices → PC, Cable, cisco server

③ connect the devices according to topology in the Manual

④ + Use copper straight - through cable blw PC and wireless router

+ copper straight - cable blw wireless routers and cable medium.

+ coaxial blw cable medium and Internet cable

+ copper straight through cable blw Internet cloud and cisco comm server.

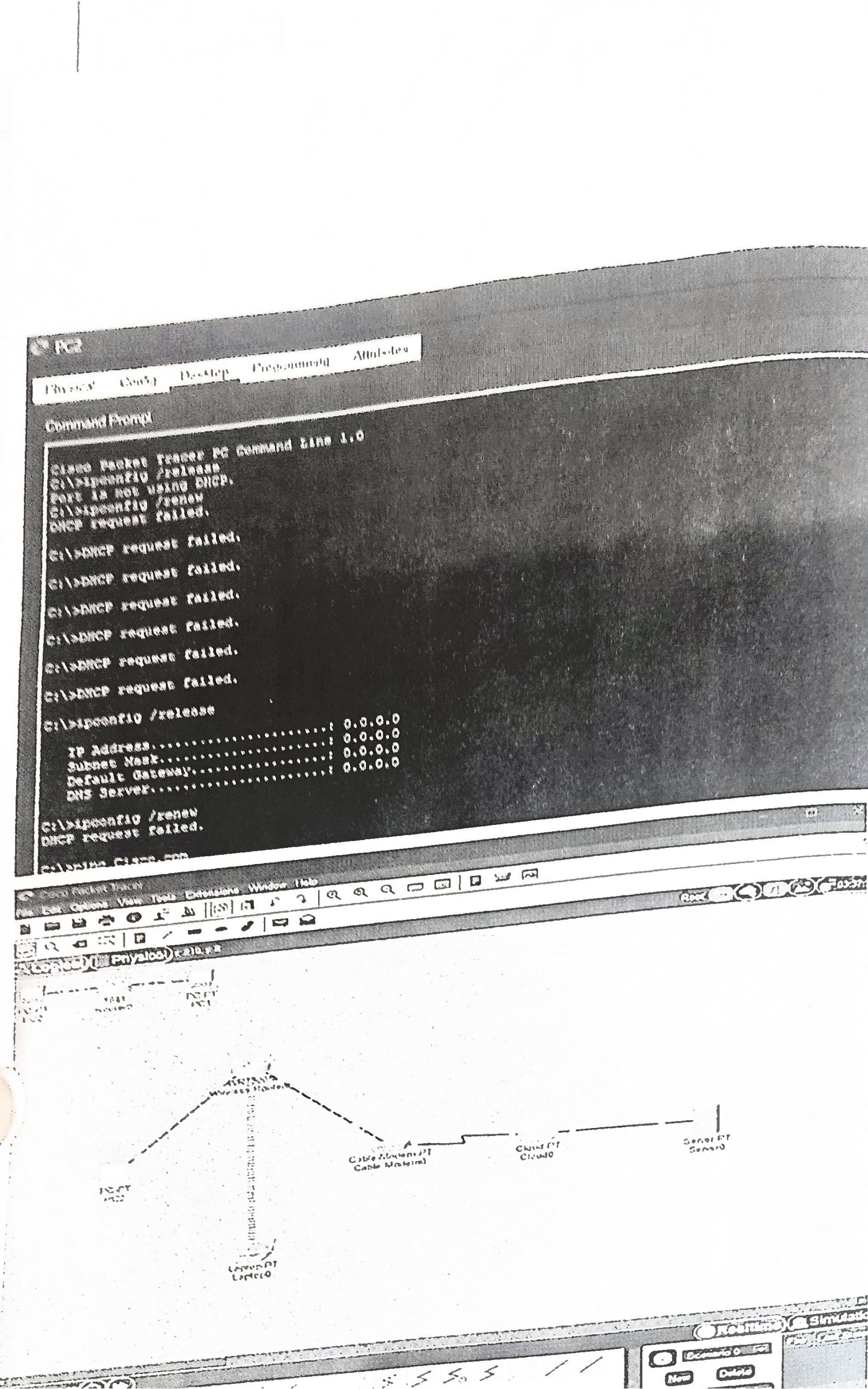
Step2: Configure wireless Routers

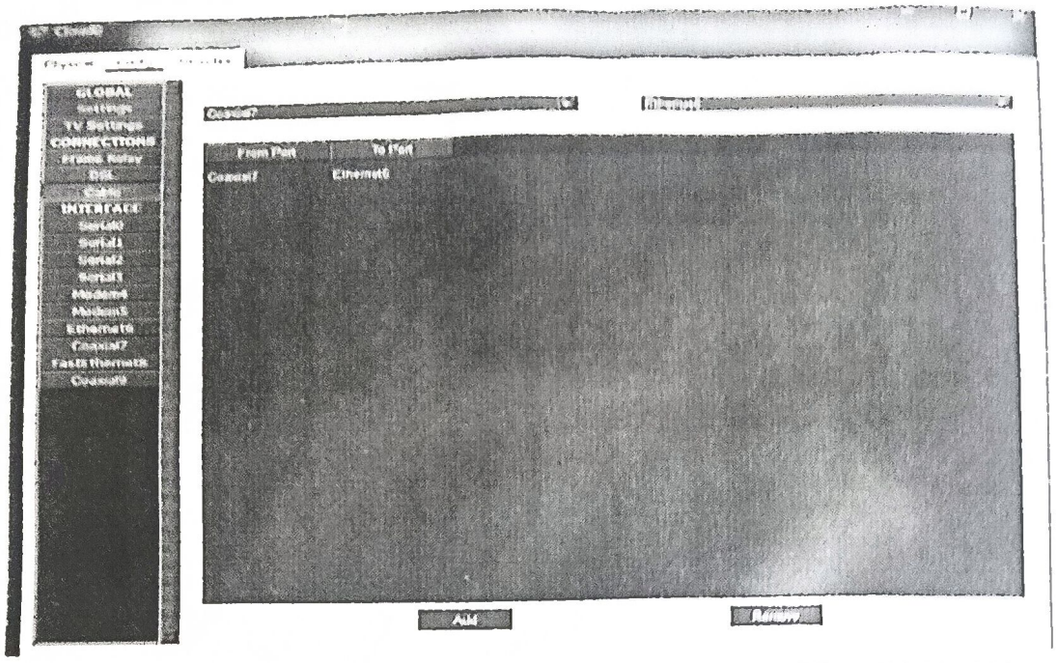
① open the wireless router, select GUI → wireless

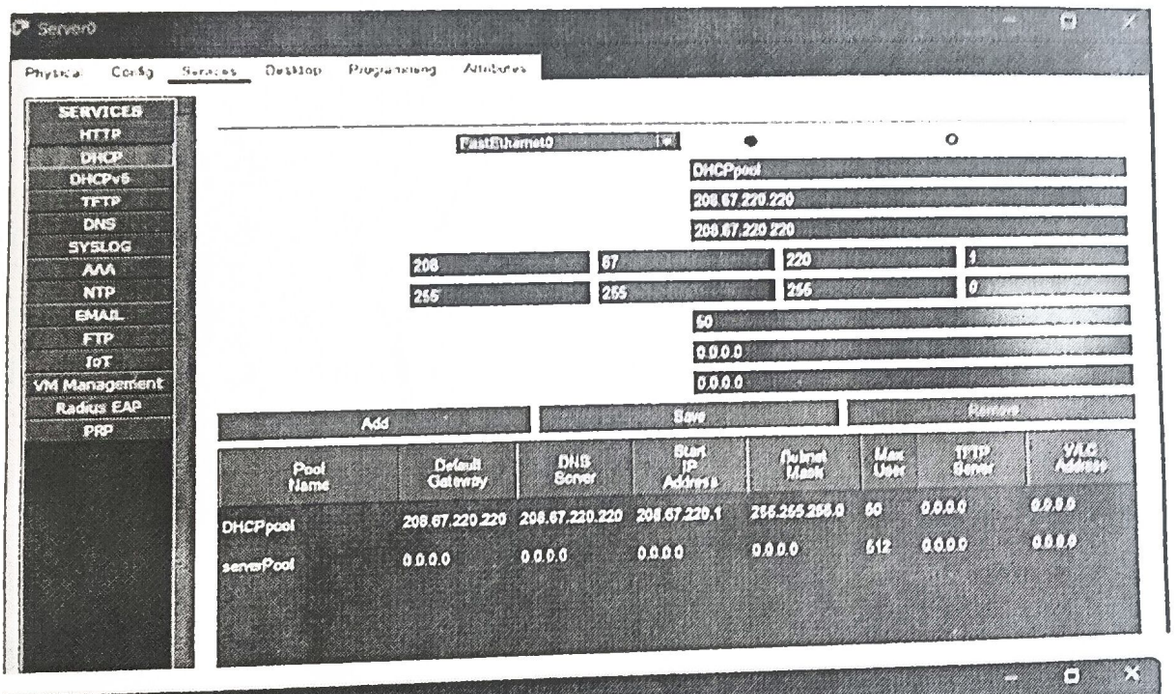
② Set the network name (SSID) to Home network

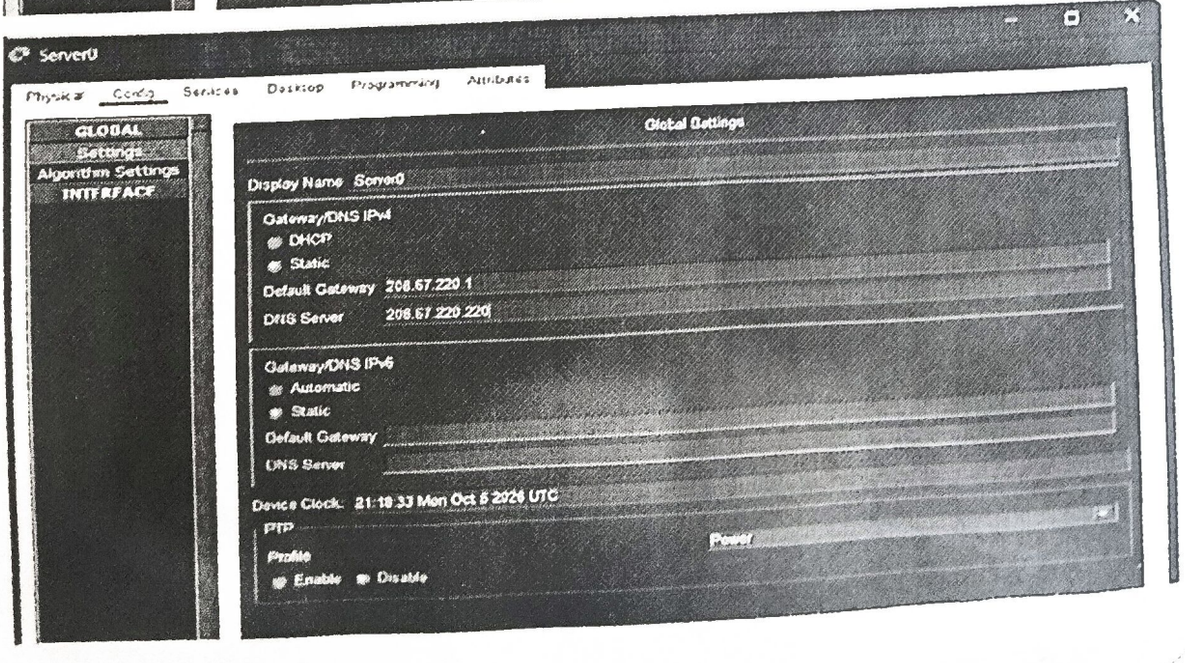
③ Configure the DNS server

④ Save the settings


PC2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /release
Port is not using DHCP.
C:\>ipconfig /renew
DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>DHCP request failed.
C:\>ipconfig /release
IP Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: 0.0.0.0
DNS Server.....: 0.0.0.0
C:\>ipconfig /renew
DHCP request failed.
Cable Modem PT
Cloud PT
Server PT
Server0


GLOBAL
INTERFACE
From Port
To Port
Add
Remove


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PPP
FastEthernet0
DHCPpool
208.67.220.220
208.67.220.220
208
67
220
1
255
255
255
0
50
0.0.0.0
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
VLAN Address
DHCPpool
208.67.220.220
208.67.220.220
208.67.220.1
255.255.255.0
50
0.0.0.0
0.0.0.0
serverPool
0.0.0.0
0.0.0.0
0.0.0.0
0.0.0.0
512
0.0.0.0
0.0.0.0

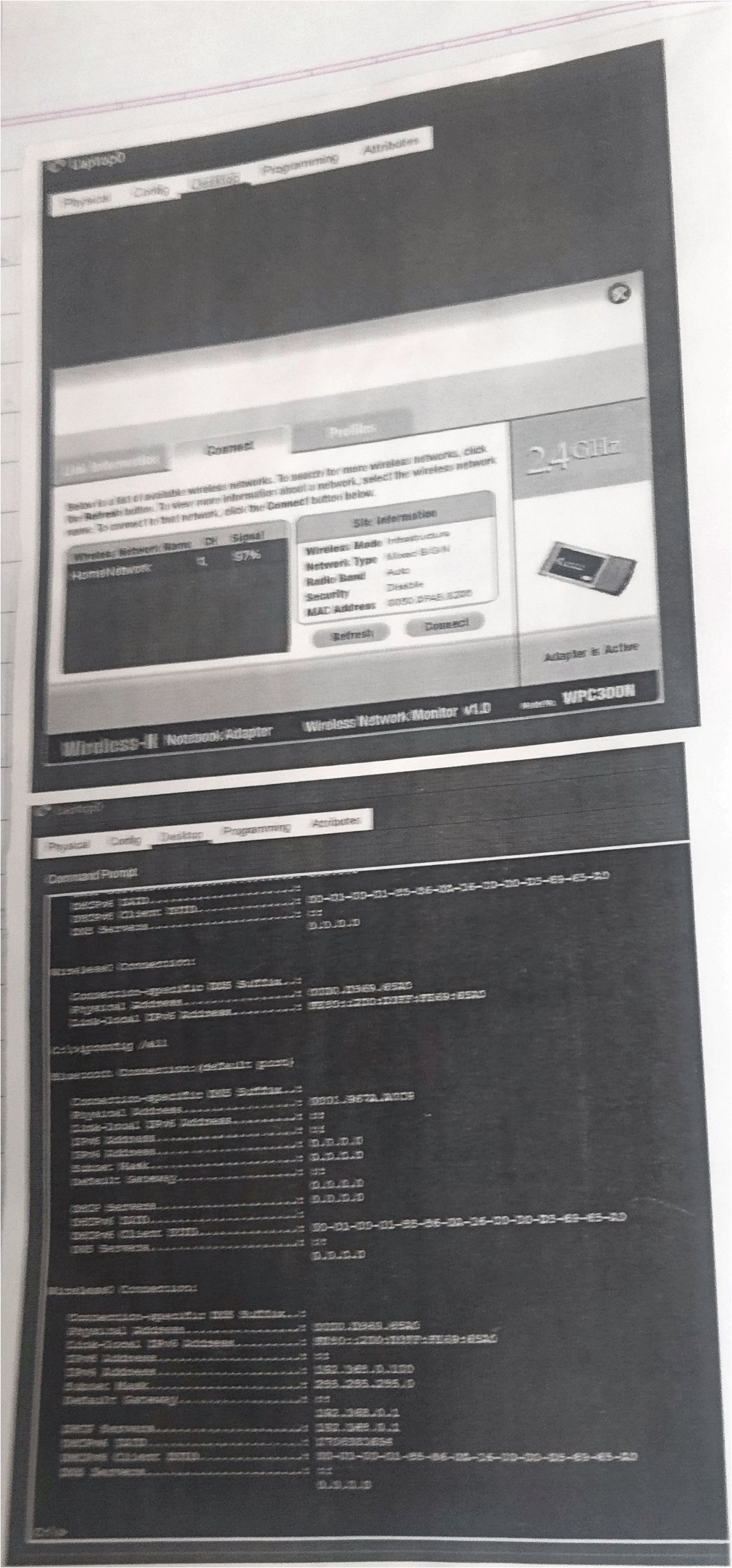

Server0
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
Global Settings
Display Name Server0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 208.67.220.1
DNS Server 208.67.220.220
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Device Clock: 21:18:33 Mon Oct 5 2026 UTC
PPP
Power
Profile
Enable
Disable

Step 3:
+ Configure the Laptop.
+ Install the wireless WPC 300N Models and Power on the Laptop.
+ Go to Desktop → PC wireless
  ↳ Select the connect tab and connect to the Homenetwork wireless network.

Step 4: configure the PC using DHCP:
+ Open the PC
+ Select Desktop → IP configuration
+ Select the DHCP option
+ The PC recieves the IPv4 configuration automatically from wireless router

Step 5: Configure cisco.com server on DHCP server
+ Open the cisco.com server
+ Select the Services tab
+ Select DHCP.
+ Turn the DHCP services on and configure the DHCP.

Step 6: Configure DNS server.
+ Select DNS server under services and turn DNS on
+ Click Name: ciscocom
  Type: A record
  Address: 208.67.220.220
+ Click Add


Laptop0
Physical
Config
Desktop
Programming
Attributes
Connect
Profiles
Link Information
2.4GHz
Below is a list of available wireless networks. To search for more wireless networks, click the Refresh button. To view more information about a network, select the wireless network name. To connect to that network, click the Connect button below.
Wireless Network Name
Ch
Signal
HomeNetwork
11
97%
Site Information
Wireless Mode
Infrastructure
Network Type
Mixed B/G/N
Radio Band
Auto
Security
Disable
MAC Address
0050.0FAE.6200
Refresh
Connect
Adapter is Active
Wireless-II Notebook Adapter
Wireless Network Monitor V1.0
Model No.
WPC300N
Laptop1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Wireless0 Connection:
Wireless1 Connection:

Step 7: Configure Fast Ethernet 0.

* Select Config → Fast Ethernet 0
* Select Static.
* Enter: IP Address: 208.67.220.20.
  Subnet Mask 255.255.255.0

Step 8: Configure DNS Server

↳ Select DNS under Services and turn DNS on.

↳ Entr Name: cisco.com
  Type: A record
  Address: 208.67.220.220

Step 9: Verify connectivity

↳ Refresh PC IPV$_H$ Setting

↳ Open the PC

* Go to Desktop → CMD Prompt.
* Enter:

  ipconfig/release
  ipconfig/renew
  ip config/all

Step 10: Test Cisco.com connectivity

Step 11: Save the file

Result:

Thus an Internetwork using a wireless router, DHCP server and Internet Cloud was successfully designed and configured in Ci packet tracer, Connectivity was success